

Day 17 – Cron Jobs & Scheduling Shell Scripts in Linux

1. What is Cron?

Cron is a Linux utility used to schedule commands or scripts to run automatically at specific times and dates.

Real-Time Examples

- Daily database backups
 - Log file cleanup
 - Server health checks
 - Automated application restarts
 - Disk utilization monitoring
 - Sending email reports
-

2. Cron Service Management

Check Cron Service Status

For RHEL/CentOS/Amazon Linux:

```
sudo systemctl status crond
```

For Ubuntu:

```
sudo systemctl status cron
```

Start Cron Service

```
sudo systemctl start crond
```

Enable Cron at Boot

```
sudo systemctl enable crond
```

Restart Cron Service

```
sudo systemctl restart crond
```

3. Understanding Crontab

A **Crontab (Cron Table)** contains scheduled tasks.

View Existing Cron Jobs

```
crontab -l
```

Edit Crontab

```
crontab -e
```

Remove All Cron Jobs

```
crontab -r
```

View Another User's Crontab

```
sudo crontab -u username -l
```

4. Cron Expression Format

```
* * * * * command
```

```
| | | | |  
| | | | └─ Day of Week (0-7)  
| | | └─── Month (1-12)  
| | └───── Day of Month (1-31)  
| └──────── Hour (0-23)  
└────────── Minute (0-59)
```

Field	Values
-------	--------

Minute	0-59
--------	------

Hour	0-23
------	------

Day of Month	1-31
--------------	------

Month	1-12
-------	------

Day of Week	0-7
-------------	-----

5. Common Cron Expressions

Every Minute

* * * * * command

Every 5 Minutes

*/5 * * * * command

Every 10 Minutes

*/10 * * * * command

Every Hour

0 * * * * command

Daily at 12:00 AM

0 0 * * * command

Daily at 6:00 AM

0 6 * * * command

Every Sunday at Midnight

0 0 * * 0 command

Every Month on 1st Day

0 0 1 * * command

Every Year on January 1st

0 0 1 1 * command

6. Special Cron Keywords

Keyword	Meaning
----------------	----------------

@reboot	Run after server reboot
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@hourly	Every hour
---------	------------

@daily	Once a day
--------	------------

@weekly	Once a week
---------	-------------

Keyword	Meaning
---------	---------

@monthly	Once a month
----------	--------------

@yearly	Once a year
---------	-------------

Examples

@reboot /home/ec2-user/startup.sh

@daily /home/ec2-user/backup.sh

7. Scheduling Shell Scripts using Cron

Step 1: Create Script

```
vi backup.sh
```

```
#!/bin/bash
```

```
DATE=$(date)
```

```
echo "Backup Started at $DATE" >> /tmp/backup.log
```

Step 2: Make Script Executable

```
chmod +x backup.sh
```

Step 3: Test Script

```
./backup.sh
```

Step 4: Configure Cron Job

```
crontab -e
```

Add:

```
*/2 * * * * /home/ec2-user/backup.sh
```

Runs every 2 minutes.

8. Redirecting Output to Log File

```
*/5 * * * * /home/ec2-user/backup.sh >> /tmp/backup.log 2>&1
```

Explanation

Symbol Meaning

>> Append output

2>&1 Redirect errors to same log

9. Cron Job Examples for DevOps Engineers

Disk Usage Monitoring

Script:

```
#!/bin/bash
```

```
df -h > /tmp/disk_usage.log
```

Cron:

```
0 * * * * /home/ec2-user/disk.sh
```

Runs every hour.

Memory Usage Monitoring

Script:

```
#!/bin/bash
```

```
free -m > /tmp/memory.log
```

Cron:

```
*/30 * * * * /home/ec2-user/memory.sh
```

Runs every 30 minutes.

Application Health Check

```
#!/bin/bash
```

```
curl http://localhost:8080
```

Cron:

```
* /5 * * * * /home/ec2-user/healthcheck.sh
```

Runs every 5 minutes.

10. Verify Cron Execution

Check Logs:

RHEL/CentOS/Amazon Linux:

```
sudo tail -f /var/log/cron
```

Ubuntu:

```
sudo tail -f /var/log/syslog
```

11. Cron Environment Variables

Check current environment:

```
env
```

Cron uses a limited environment.

Use full paths:

```
/usr/bin/java
```

```
/usr/bin/python3
```

```
/bin/bash
```

Example:

```
* /5 * * * * /bin/bash /home/ec2-user/script.sh
```

12. Common Cron Troubleshooting

Permission Issues

`chmod +x script.sh`

Incorrect Path

Use full path:

`which java`

`which python3`

Cron Service Not Running

`systemctl status crond`

Script Running Manually but Not in Cron

Use:

`#!/bin/bash`

and absolute paths.

Hands-On Lab

Task 1

Create a script to display current date and time.

Schedule it every minute.

Task 2

Create a script that writes disk usage to a file.

Schedule it every 10 minutes.

Task 3

Create a script that checks whether Tomcat is running.

Schedule it every 5 minutes.

Task 4

Create a backup script for /etc.

Schedule it daily at midnight.

Interview Questions

1. What is Cron?
2. What is Crontab?
3. How do you edit a Crontab?
4. Explain the Cron expression format.
5. Difference between @daily and 0 0 * * *.
6. How do you verify Cron job execution?
7. Why do Cron jobs fail?
8. How do you redirect Cron output to a log file?
9. What is 2>&1?
10. How is Cron used in DevOps projects